

WASP-1b

R-2018

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_121217 · created 2026-08-07T23:48:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456278.7526 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.014
Transit depth (Rp/Rs)^2	0.98 +/- 0.27 [%]
Orbital Inclination (inc)	89.82 +/- 1.42
Airmass coefficient 1 (a1)	0.9884 +/- 0.0068
Airmass coefficient 2 (a2)	0.0115 +/- 0.0066
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	#2 - [202, 368]
Optimal Aperture	4.44
Optimal Annulus	10.15
Transit Duration (day)	0.098 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.099 $\pm$ 0.014 vs 0.1036 (deviation 0.32 σ)
Duration fit vs published	0.098 d vs 0.1567 d (deviation 14.58 σ)
Mid-time O-C	11.6 min
Depth SNR	6.9
Residual scatter	0.68 %

Light curve

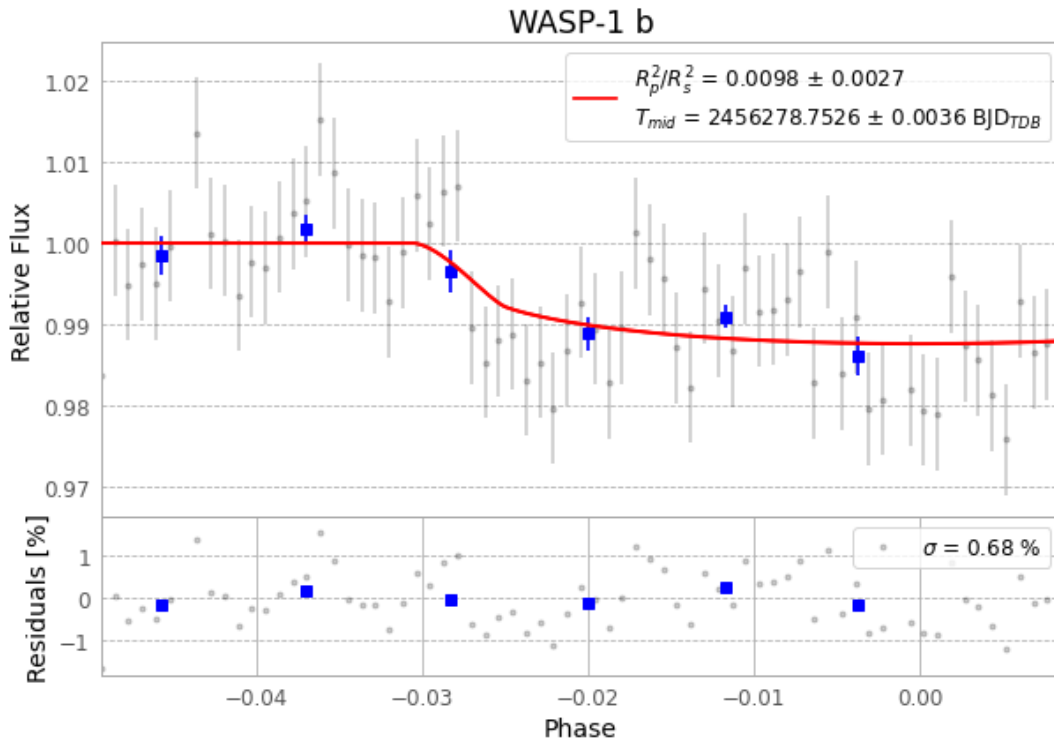


Figure 1 — FinalLightCurve_WASP-1 b_2012-12-16.png (SHA-256 83556d8ff854db04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 313], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 58447e669838b3b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

71 FITS frames (47 MB), WASP-1121217030012.FITS → WASP-1121217063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121217.log (c818d468cc0f...), FinalLightCurve_WASP-1 b_2012-12-16.png (83556d8ff854...), FinalLightCurve_WASP-1 b_2012-12-16.csv (619970a7461e...)

Compute resources

Wall time 131.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:48Z.